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2. A class of year 11 boys were investigating reaction time. The students suggested the following hypothesis:

"Year 11 students have faster reaction times than teachers"

A computer program was used to record the reaction time. Each individual had to press a button on the keyboard when the screen turned green (**Image 2.1**). Each individual had three attempts and the mean value was recorded.

Image 2.1

(a) (i) State the stimulus and the receptor involved in this investigation. [2]

Stimulus: ....

Receptor: ....

(ii) Describe how the information travels from the **receptor** to the **central nervous system**. [2]

....

....

....

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(b) The results for the students are shown in **Table 2.2** and the teachers in **Table 2.3**.

Table 2.2

Name	Age	Reaction time (ms)
Rhidian	15	382
Iestyn	15	412
Reuben	15	375
James	15	399
Harvey	15	401
		Mean reaction time = 394

Table 2.3

Name	Age	Reaction time (ms)
Miss Williams	42	479
Mr Davies	32	391
Mrs Wilcox	37	415
Mr Jones	55	475
Mrs Evans	48	431
		Mean reaction time = ...

(i) Complete **Table 2.3** by calculating the mean reaction time for the teachers **to the nearest whole number**. [2]
Space for working.

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(ii) Evaluate the extent to which the results in **Tables 2.2** and **2.3** support the students' hypothesis. [2]

You should do this by giving:
- **one** piece of evidence that supports the hypothesis
- **one** piece of evidence that does not support the hypothesis

Evidence that supports hypothesis

....

....

Evidence that does not support the hypothesis

....

....

(iii) State **one** variable that should have been controlled in this investigation. [1]

....

(iv) State **one** way that the students could have increased their confidence in their results. [1]

....

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